Group Project
An important part of this course is a group project

Description

Your project is a key part of the course. You need to do it in a group with maximum 5 students and minimum 2 students. Each group will design and implement a good Machine Learning (ML) Project addressing at least one good application (a sample list of applications is shown below). Because ML is an important part of AI and AI is growing fast, you are encouraged to do an Intelligent that will use a good ML algorithm in your project i.e. your project can use an Intelligent Agent that will use ML as the key learning component.

A good project is one that applies one or more of ML algorithms covered in class, and in some good ways – e.g. in design, implementation, using datasets, Knowledge Base (If includes an AI Agent). An excellent project is a project where you have shown some novelties in the Intelligent Agent / ML Model – Architecture, algorithms, Agentic part.

The project will provide you with a unique opportunity for exploring one or more areas of ML/AI. You can pick some algorithms that we did not cover or did not cover in depth in class. Some examples can be from Neural Net, Deep Learning, Transformers, AI, Generative AI, Logic, Cognitive Computing or their combinations. It is highly recommended to design a good ML system, preferably using Intelligent Agent(s) as a key part of your project addressing some real world problems.

Your project, in general, will need some datasets. You should choose appropriate datasets, apply AI / Machine Learning algorithms from above mentioned fields in your Agent. Then analyze / evaluate results, and make recommendations / future work. Pick a project that you are excited about. I will closely work with you throughout the whole process!

Due to the growing successes of AI, Gen AI, Reinforcement Learning, Neuro-Symbolic Learning and the like, you may select a topic from these areas as well if you are interested and comfortable. I will be happy to help you in these areas. It can help you better prepare to be attracted in the industry.

If you want to tie the class project to your research project or a fun application, you are strongly encouraged to do so. In this case you can use algorithms that we have covered in the class or any other algorithms. However, you should be able to demonstrate some novelty. Simply applying an algorithm to a sub-problem in your research project / application is not acceptable. Some novelty is important for all projects to earn higher points.

Project Deliverables

In a nutshell

- Initial proposal (3rd week). 10% of your project grade – see the format below. You may take a few more days – but final version by 5-th week.
- A video and / or slides of your talk on week 15, 45% of your project grade.
- A paper describing your work (16th week). 25% of your project grade.
- Code and scripts (16th week). 20% of your project grade.

Project proposal format: Proposals should be one page maximum. Include the following information:

- Project title
- Data set and tools / platforms
- Project idea. This should be approximately two paragraphs.
- Platform / tools / programming language you will need to use.
- Papers to read. Include 1-3 relevant papers. You will probably want to read at least one of them before submitting your proposal.
- Teammates: Minimum team size 2, Maximum team size is 5 students. I do not recommend a single person project as it will be too much work!
- 15^{th} week milestone: What will you complete by 15^{th} week? Good results are expected during your presentation.

Sample Project Topics

Consider using sample project topics from the following list (you can also select from other similar topics e.g. from searching on the Internet):

Recommended to use AI / Agentic AI. Also can use Co-Pilots BUT with your additional work – additional work will be the main focus to evaluate your projects).

1. Heart Disease Prediction
2. Hand written digit or character recognition
3. Healthcare Agent improving Personalized Health Care
4. A Digital Twin in manufacturing (e.g. shop floor performance improvements)
5. Customer Support Agent
6. Work Flow Automation ………
7. Fraud Detection / Prevention
8. Customer Recommendations
9. Chatbots with effective Conversations (including LLM based bots)
10. Facial Emotion Recognition and Detection
11. Online Assignment Plagiarism Checker
12. Personality Prediction System via CV Analysis
13. Heart Disease Prediction

14. Differentiate the music genre from an audio file
15. Image reconstruction by using an occluded scene
16. Identify human emotions through pictures
17. Summarize articles written in technical text
18. Drawing inference from a document
19. Content Generation – for various applications
20. Other Similar applications of your choice